

17 Oscillations

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Simple harmonic motion has acceleration proportional to displacement and opposite in direction: $a = -\omega^2 x$. A mass-spring system has $T = 2\pi\sqrt{m/k}$ when damping is negligible. At equilibrium speed is maximum; at extreme displacement speed is zero. Resonance occurs when driving frequency is close to natural frequency.

Worked example: follow the reasoning

If $\omega = 4 \text{ rad s}^{-1}$ and displacement $x = 0.030 \text{ m}$, acceleration $a = -16 \times 0.030 = -0.48 \text{ m s}^{-2}$.

Quick check · multiple choice

A Which expression describes SHM acceleration?

[1]

A $a = -\omega^2 x$

B $a = +\omega^2 x$

C $a = \omega x^2$

D $a = 0$ throughout

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A spring oscillator has mass 0.50 kg and $k = 200 \text{ N m}^{-1}$. Find period and describe the effect of quadrupling mass.

[3]

Final answer: _____

Answers and explanation

Quick check: A. $a = -\omega^2 x$

Question 1 [3 marks]

$T = 2\pi\sqrt{m/k} = 2\pi\sqrt{0.50/200} \approx 0.314 \text{ s}$ [2]. Quadrupling m doubles T [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.